

Shreya Uprety

AI Engineer

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Experience

AI Engineer Jan 2026 – Present
MedforceAI, GenAI Clinic OS Remote (London, UK)

- Build AI systems for medical pre-consultation, and led the migration from a monolithic backend to an event-driven, multi-agent architecture on Google Cloud so features ship independently.
- Built and optimized chat and voice agents for medical terminology transcription and clinical interpretation, adding retrieval and verification layers that reduced confident but incorrect answers.
- Own the FastAPI backend, the retrieval stack (FAISS and ChromaDB), and the Dockerized deployment pipeline.
- Authored technical documentation and material for investor and stakeholder presentations.

AI Fellow 2025
Fusemachines, AI Fellowship

- Completed a year-long applied machine learning fellowship with industry mentorship and weekly project reviews.
- Built NLP and deep learning pipelines end to end, from data preparation through model evaluation.

Education

Bachelor of Engineering in Computer Engineering 2022 – 2026
Tribhuvan University, Institute of Engineering, Pulchowk Campus

Technical Skills

- **Languages:** Python, JavaScript, SQL
- **AI / ML:** PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, LLaMA, YOLOv8, BLIP, Ollama, RAG, NLP, Computer Vision
- **Web & Backend:** FastAPI, Django, Flask, React.js, Next.js, Tailwind CSS
- **Data & Infra:** PostgreSQL, MySQL, SQLAlchemy, FAISS, ChromaDB, Docker, Git, Google Cloud Platform

Projects

Medical Question Answering System github.com/shreyauprety/medical-qa-system
Multi-stage RAG, LangChain and LangGraph

- Built a multi-stage RAG pipeline with three-tier retrieval (BM25, FAISS, and concept-based) and hybrid reasoning.
- Integrated UMLS concept expansion, confidence calibration, and a safety verification module that refuses ungrounded answers.

LegalDoc-RAG, Legal Document Analysis github.com/shreyauprety/legaldoc-RAG
RAG, NLP, Full-stack

- Built a RAG system with hybrid BM25 and FAISS retrieval and a LoRA-tuned LLaMA for legal reasoning.
- Shipped a full-stack app: React.js and Tailwind frontend over a FastAPI backend with a SQLAlchemy ORM.

Plant Disease Detection System github.com/shreyauprety/plant-disease-detection
Computer Vision, Transfer Learning

- Built a multi-crop disease classifier across 34 classes and 8 crop species using ResNet50, EfficientNet-B0, and a Vision Transformer.

StockSage, Financial Analytics Platform github.com/shreyauprety/StockSage
Machine Learning, Sentiment Analysis

- Built forecasting dashboards with news sentiment analysis for the Nepali stock market using React.js and Django.

Leadership

CIT Head, Digital Literacy Drive Nov 2025
WorldLink Communications and LOCUS, Sudurpaschim Province, Nepal

- Led digital literacy workshops on internet safety and cybersecurity, reaching over 385 students.

Certifications

- **Fusemachines AI Fellowship** 2025
- **Associate Data Scientist in Python**, DataCamp 2023